

Manojkumar Kumaran

Computational Biologist • Bioinformatics & Neuroscience

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RESEARCH SUMMARY

Bioinformatics researcher with 10+ years of experience in mammalian genomics and next-generation sequencing analysis, and a Ph.D. in Bioinformatics. Author of 17 publications (10 peer-reviewed, 6 as first author) spanning genomics, ophthalmic genetics, cancer, and neuroscience, with current work in the computational neuroscience of spinal-cord injury and axon regeneration. Builds and benchmarks analytic pipelines for single-cell RNA/ATAC and whole-exome data and machine-learning tools for variant prioritization and behavioural analysis. Seeking to present this work and expand collaborations in computational neuroscience.

EDUCATION

Ph.D., Bioinformatics 2015 – 2022
SASTRA University

M.Tech., Bioinformatics (Integrated) 2009 – 2015
Bharathidasan University

RESEARCH EXPERIENCE

Research Associate Oct 2022 – Present
CSIR – Centre for Cellular & Molecular Biology (CCMB)

- Developed automated pipelines for single-cell RNA, single-cell ATAC, and bulk ATAC sequencing analysis.
- Built a machine-learning model for automated mouse behavioural analysis.

Data Scientist Aug 2021 – Sep 2022
INDX Technologies, Mumbai

- Developed an automated tool for analysing whole-exome sequencing (WES) data from cancer patients.
- Built an automated variant-annotation tool integrating available drug-response information.

Junior Research Fellow Aug 2015 – Aug 2021
Aravind Medical Research Foundation

- Developed and benchmarked analytic tools and pipelines for next-generation sequencing (NGS) data.
- Applied biostatistics and bioinformatics algorithms to integrate multi-level biological information with functional genomic data for hypothesis generation, variant identification, and variant prioritization.
- Collaborated across disciplines and contributed to multiple organisational projects.

PEER-REVIEWED PUBLICATIONS

12 publications, 7 as first author.

1. Krishna N, **Kumaran M**, et al. PATZ1 remodels the nucleosome landscape to promote chromatin accessibility in injured neurons. *Communications Biology* (2026).
2. **Kumaran M**, et al. Deep Neurite Analysis Tool (DeNAT): A machine-learning framework for precise automated neurite outgrowth measurement. *BMC Bioinformatics* (2025). *[First author]*
3. John ASS, **Kumaran M**, et al. Genomic analysis identifies unreported adaptive mutations in monkeypox virus immune evasion genes. *Virus Genes* (2026).
4. **Kumaran M**, Wang Z, Batsel E, et al. Single-nuclei sequencing reveals a robust corticospinal response to nearby axotomy but overall insensitivity to spinal injury. *J Neurosci* (2025). *[First author]*

5. Jeyaprakash K, **Kumaran M**, et al. Investigating druggable kinases for targeted therapy in retinoblastoma. *J Hum Genet* (2024).
6. Shafi G, **Kumaran M**, et al. Comprehensive ctDNA profiling reveals potential resistance mechanisms. *Cancer Research* (2024).
7. **Kumaran M**, et al. eyeVarP: A computational framework for identification of pathogenic variants specific to eye disease. *Medical Genetics* (2023). *[First author]*
8. **Kumaran M**, et al. Clinical reassessments and whole-exome sequencing. *Ophthalmic Genetics* (2021). *[First author]*
9. **Kumaran M**, et al. Bioinformatics for whole-exome studies. *Bioinformatics and Human Genomics Research* (2021). *[First author]*
10. **Kumaran M**, et al. Whole-exome sequencing identifies multiple pathogenic variants. *Indian Journal of Ophthalmology* (2021). *[First author]*
11. Raj RK, **Kumaran M**, et al. Genetic characterization of Stargardt clinical cases. *Eye and Vision* (2020).
12. **Kumaran M**, et al. Performance assessment of variant calling pipeline using human whole-exome sequencing and simulated data. *BMC Bioinformatics* (2019). *[First author]*

PREPRINTS & MANUSCRIPTS UNDER REVIEW

1. **Kumaran M**, et al. KiMA: Kinematic Motion Analysis for Spinal Cord Injury Research. *bioRxiv* (2026). *[First author]*
2. Sahu Y, **Kumaran M**, et al. Combinatorial Control of Corticospinal Axon Growth by Retinoic Acid Receptors. *bioRxiv* (2026).
3. Menon AS, **Kumaran M**, et al. Rewiring of the three-dimensional genome encodes regenerative potential in the adult central nervous system. *bioRxiv* (2026).
4. Sanyal K, **Kumaran M**, et al. A Chemically Stable Retinoic Acid Mimic Drives Regeneration and Behavioral Recovery after Spinal Cord Injury. *bioRxiv* (2025).
5. Menon AS, **Kumaran M**, et al. PATZ1 reinstates a growth-permissive chromatin landscape in adult corticospinal neurons after injury. *bioRxiv* (2025).
6. Sahu Y, **Kumaran M**, et al. Nuclear receptor transcription factors promote axon regeneration in the adult corticospinal tract. *bioRxiv* (2025).

CONFERENCE PRESENTATIONS

- **Poster —** Indian Academy of Neuroscience (IAN), 2026. *[Oral Presentation]*
- **Poster —** K-brain, 2026. *[Travel Award]*
- **Poster —** Cellular & Molecular Mechanisms of Development and Regeneration (CMMDR), 2026. *[Best Poster]*
- **Poster —** Indian Academy of Neuroscience (IAN), 2024. *[Travel Award]*

AWARDS & HONOURS

- **Travel Award** — K-brain, 2026
- **Best Poster Award** — Cellular & Molecular Mechanisms of Development and Regeneration (CMMDR) Conference, 2026.
- **Travel Award** — Indian Academy of Neuroscience (IAN), 2024
- **Best Poster Award (VRM Enrolment Award)** — Aravind Medical Research Foundation, 2019.
- **Junior Research Fellowship** — Aravind Medical Research Foundation, 2015–2021.

TECHNICAL SKILLS

Programming: R, Python, Shell (10+ years; tool/pipeline development for NGS analysis).

Genomics & NGS: Single-cell RNA/ATAC & bulk ATAC analysis, gene-expression analysis and visualization, pathway & network analysis, sequence annotation, variant calling and prioritization, machine learning for pathogenic-variant filtering.

Other: Nucleic-acid & protein sequence/structural analysis; Windows, Unix/Linux & macOS (10+ yrs); systems hardware, networking and server administration (6 yrs); MS Office; Adobe Photoshop, Illustrator & Lightroom.